Solution:

Here, the event of interest is selecting a female with age above 25 years. In connection with the occurrence of this event, the following two events must happen.

A: a person selected is female

B: a person chosen is above 25 years.

Situation1:

We are interested in the event B, given that A has occurred. This event can be denoted by $B|A$. It can be read as "B given A". It means that first the event A occurs then under that condition, B occurs. Here, we want to find the probability for the occurrence of $B|A$ i.e., $P(B|A)$. This probability is called conditional probability. In reverse, the probability for selecting a female given that a person has been selected with age above 25 years is denoted by $P(A|B)$.

Situation 2:

Suppose that it is interested to select a person who is both female and with age above 25 years. This event can be denoted by $A \cap B$.

Calculation of probabilities in these situations warrant us to have another theorem namely Multiplication theorem. It is derived based on the definition of conditional probability.

$$P(A) = P(\text{Selecting a female}) = \frac{1500}{4000}$$

$$P(A \cap B) = P(\text{Selecting a female with age above 25 years}) = \frac{400}{1500}$$

Hence, $$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{400}{1500} \times \frac{4000}{1500} = \frac{160}{225} = \frac{32}{45}.$$

8.5.1 Definition of Conditional of Probability

If $P(B) > 0$, the conditional probability of A given B is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad .$$

If P(B) = 0, then $P(A \cap B) = 0$. Hence, the above formula is meaningless when $P(B) = 0$. Therefore, the conditional probability P(A|B) can be calculated only when P(B)>0.

The need for the computation of conditional probability is described in the following illustration.